

$$\underline{\text{Derm}}: f(x) = \sum_{k=0}^{\infty} \frac{(x-a)^k}{k!} f^{(k)}(a) + \frac{1}{m!} \int_a^x (x-t)^m f^{(m+1)}(t) dt$$

$$= T_m(f)(x) + \frac{1}{m!} \int_a^b (x-t)_+^m f^{(m+1)}(t) dt$$

$$A(f) = \underbrace{A(T_m(f))}_{=0} + \frac{1}{m!} \int_a^b A_x [(x-t)_+^m] f^{(m+1)}(t) dt$$

Corolar: Dacă A este liniară și are gradul de exactitate m și dacă $A'_x(x-t)_+^m$ păstrează semn constant, atunci

$\exists \theta \in (a, b)$ a.t.:

$$A(f) = \frac{1}{m!} f^{(m+1)}(\theta) \int_a^b A_x (x-t)_+^m dt$$



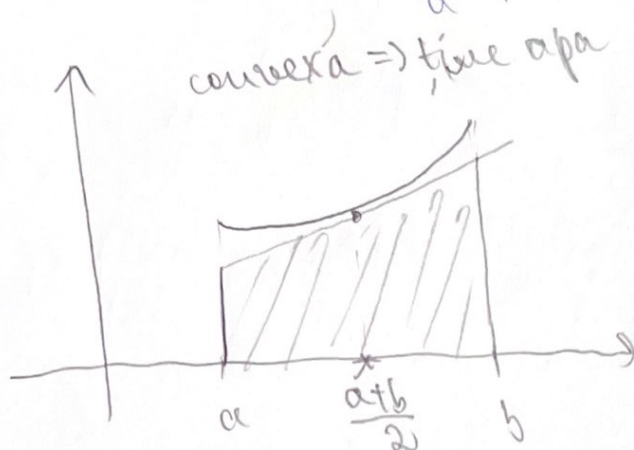
Formula dreptunghiului:

$$\int_a^b f(x) dx = (b-a) f\left(\frac{a+b}{2}\right) + R(f)$$

$$R(f) = \frac{(b-a)^3}{24} f''(\theta)$$

$$f \text{ convexă} \Rightarrow \int_a^b f(x) dx \geq (b-a) f\left(\frac{a+b}{2}\right)$$

$$f \text{ concavă} \Rightarrow \int_a^b f(x) dx \leq (b-a) f\left(\frac{a+b}{2}\right)$$



$$\int_a^b f(x) dx \geq (b-a) f\left(\frac{a+b}{2}\right)$$

Formula dreptunghiurilor

$$[a, b], \Delta_n: a = x_0 < x_1 < \dots < x_k < x_{k+1} < \dots < x_n = b$$

$$x_k = a + b \cdot \frac{b-a}{n}$$

$$\int_a^b f(x) dx = \sum_{k=0}^{n-1} \int_{x_k}^{x_{k+1}} f(x) dx = \sum_{k=0}^{n-1} \left[\frac{b-a}{n} f\left(a + \frac{b-a}{2n}(2k+1)\right) + \frac{(b-a)^3}{24n^3} f''(\theta_k) \right]$$

$$= \frac{b-a}{n} \sum_{k=0}^{n-1} f\left(a + \frac{b-a}{2n}(2k+1)\right) + \frac{(b-a)^3}{24n^2} \cdot \left[\sum_{k=0}^{n-1} \frac{f''(\theta_k)}{n} \right]$$

$\theta_k \in [x_k, x_{k+1}]$

proprietatea lui Darboux:

$$\exists \theta \in [a, b] \text{ a.î. } \sum_{k=0}^{n-1} \frac{f''(\theta_k)}{n} = f''(\theta)$$

$$\int_a^b f(x) dx = \frac{b-a}{n} \sum_{k=0}^{n-1} f\left(a + \frac{2k+1}{2n}(b-a)\right) + \frac{(b-a)^3}{24n^2} f''(\theta)$$

$$\|f''\| \leq M, \quad R_n(f) \leq \frac{(b-a)^3}{24n^2} M < \varepsilon$$

$$\int_0^1 e^{-x^2} dx \quad \text{aproximare la 2 zecimale}$$

lungime interval: $\frac{1}{n}$

$$x_k = \frac{2k+1}{2n}$$

$$e^{-\left(\frac{2k+1}{2n}\right)^2}$$

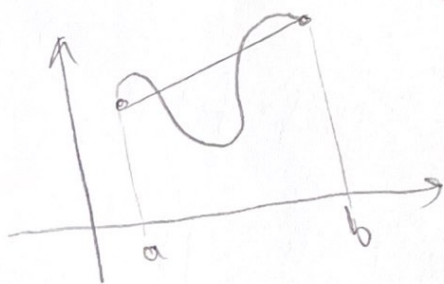
$$f'(x) = 2xe^{-x^2}$$

$$|f''(x)| = |-2e^{-x^2} + 4x^2e^{-x^2}| = |2e^{-x^2}(2x^2-1)| \leq 2$$

$$\frac{1}{24n^2} \cdot 2 \leq \frac{1}{100} \quad \Rightarrow \quad 3n^2 \geq 25$$

$$\boxed{n=9}$$

Formula trapezului



$$\int_a^b f(x) dx = \frac{f(a) + f(b)}{2} (b-a) + R(f)$$

$$R(f) = \int_a^b [f(x) - L(f; a, b)(x)] dx$$

$$= \int_a^b \underbrace{(x-a)(x-b)}_{\text{nu schimba semn}} \cdot [x, a, b; f] dx = \frac{h(\theta)}{h'(\theta)} \cdot \int_a^b (x-a)(x-b) dx$$

$f \in C^2[a, b] \quad h(\theta) = [\theta, a, b; f]$

$$\exists c \in (a, b) \text{ a. i. } h(\theta) = \frac{f''(\theta)}{2!}$$

$$R(f) = \frac{f''(\theta)}{2!} \cdot \int_a^b (x-a)(b-x) dx$$

$$\int_a^b f(x) dx = (b-a) \frac{f(a) + f(b)}{2} + \frac{f''(\theta)}{2}$$

$$f(x) = (x-a)^2$$

$$\int_a^b (x-a)^2 dx = \frac{(b-a)^3}{3} = \frac{(b-a)^2}{2} \cdot \frac{b-a}{2} + \frac{2b}{2}$$

$$R = \frac{(b-a)^3}{3} - \frac{(b-a)^3}{2} = -\frac{(b-a)^3}{6}$$

$$\int_a^b f(x) dx = \frac{b-a}{2} (f(a) + f(b)) - \frac{(b-a)^3}{6} f''(\theta)$$

Corolar: Dacă f este o funcție convexă pe $[a, b]$, atunci: $\int_a^b f(x) dx \leq \frac{b-a}{2} (f(a) + f(b))$

$$f \text{ convexă: } \int_a^b f(x) dx \leq \frac{b-a}{2} (f(a) + f(b))$$

$$f \text{ concavă: } \int_a^b f(x) dx \geq (b-a) \left(f\left(\frac{a+b}{2}\right) \right)$$

$$(b-a) f\left(\frac{a+b}{2}\right) \leq \int_a^b f(x) dx \leq \frac{b-a}{2} (f(a) + f(b))$$

Formula trapezului

Fie $f \in C^2[a, b]$ și fie Δ_n o diviziune echidistantă, $x_k = a + k \frac{b-a}{n}$, $k = 0, n$

Additivitatea integralei față de interval.

$$\begin{aligned} \int_a^b f(x) dx &= \sum_{k=0}^{n-1} \int_{x_k}^{x_{k+1}} f(x) dx = \frac{b-a}{2n} \sum_{k=0}^{n-1} [f(x_k) + f(x_{k+1})] \\ &\quad - \frac{f''(\theta_k) \cdot (b-a)^3}{6n^3} \\ &= \frac{b-a}{2n} \left[f(a) + f(b) + 2 \sum_{k=1}^{n-2} f(x_k) \right] - \frac{b-a}{6n^2} \cdot \frac{\sum_{k=0}^{n-1} f''(\theta_k)}{n} \end{aligned}$$

$$\Rightarrow \int_a^b f(x) dx = \frac{b-a}{2n} \left[f(a) + \sum_{k=1}^{n-2} f(x_k) + f(b) \right] - \frac{(b-a)^3}{6n^2} f''(\theta)$$

$$\int_a^b f(x) dx \leq \frac{b-a}{n} \left[f(a) + \sum_{k=1}^{n-2} f(x_k) + f(b) \right]$$

$$\int_a^b f(x) dx \geq \frac{b-a}{n} \sum_{k=0}^{n-1} \frac{f(x_k + x_{k+1})}{2}$$

OPERATORI LINIARI ȘI POZITIVI DE APROXIMARE

Weierstrass: P_n , $\lim_{n \rightarrow \infty} \|f - P_n\|_{\infty} = 0$

S.N. Bernstein

$L_n \quad \forall f \geq 0 \Rightarrow L_n f \geq 0$
↑
operator
liniar

Se notează: $B_n(f)(x)$

$f \in C[0,1]$

$$B_n(f)(x) = \sum_{k=0}^n b_{n,k}(x) f\left(\frac{k}{n}\right)$$

$$b_{n,k}(x) = \binom{n}{k} x^k (1-x)^{n-k}, \quad k = \overline{0, n}$$

$$e_i(x) = x^i, \quad x \in [0,1], \quad i \in \mathbb{N}$$

$$\boxed{B_n(e_0)(x) = \sum_{k=0}^n b_{n,k}(x) = 1}$$

$$= \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k} = (x + 1 - x)^n = 1$$

de la Matematici
Speciale

$$B_n(e_1)(x) = \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k} \frac{k}{n}$$

$$= x$$

$$(1-x+tx)^n = \sum_{k=0}^n \binom{n}{k} t^k x^k (1-x)^{n-k} \quad \left| \frac{d}{dt} \right.$$

$$\frac{d}{dt} (1-x+tx)^n = \sum_{k=0}^{n-1} k t^{k-1} b_{n,k}(x) \quad \left| \cdot \frac{1}{n} \right.$$

$$x (1-x+tx)^{n-1} = \sum_{k=0}^n \frac{k}{n} t^{k-1} b_{n,k}(x) \quad | t=1$$

$$\Rightarrow \boxed{B_n(e_1)(x) = x}$$

$$B_n(e_2)(x) = n x t (1-x+tx)^{n-1} = \sum_{k=0}^n k t^k b_{n,k}(x) \quad \left| \cdot \frac{1}{t} \right.$$

$$= nx + n(n-1)x^2 = \sum_{k=0}^n k^2 b_{n,k}(x) \quad \left| \cdot \frac{1}{n} \right.$$

$$\boxed{B_n(e_2)(x) = x^2 + \frac{x(1-x)}{n}}$$

$$B_n(e_0)(x) = 1$$

$$B_n(e_1)(x) = x$$

$$B_n(e_2)(x) = x^2 + \frac{x(1-x)}{n}$$

$$n \rightarrow \infty \quad B_n(e_2) \Rightarrow x^2 ; |B_n(e_2)(x) - x^2| \leq \frac{1}{n}$$

Teoremă (Popoviciu-Bohman-Korovkin) :

$L_n: C[a,b] \rightarrow C[a,b]$. Condiția necesară și suficientă pt ca $\forall f \in C[a,b], L_n(f) \Rightarrow f$ este ca: $L_n e_i \Rightarrow e_i, i=0,1,2$.

Pt a testa că are și de operatori liniari și pozitivi, arătăm că este suficient să calculăm $B_n(f)(x)$ pt $0,1,2$.

Dăm: $f \in C'[a,b]$

$$|f(x) - f(t)| = |(x-t)f'(t)| \leq |x-t| \|f'\|_\infty$$

$$L_n(|f(x) - f(t)|)_t \leq \|f'\| L_n(|x-t|)_t$$

$$(f \leq g \Rightarrow g - f \geq 0)$$

$$L_n(g-f)(x) \geq 0$$

$$L_n g(x) - L_n f(x) \geq 0$$

$$L_n(|f(x) - f(t)|)_t \geq |L_n(f)(x) - f(x) L_n(e_0)(x)|$$

$$-|f(x) - f(t)| \leq f(x) - f(t) \leq |f(x) - f(t)|$$

$$-L_n|f(x) - f(t)|_x \leq f(x) \cdot (L_n e_0)_x - L_n f(t) \leq L_n|f(x) - f(t)|_x$$

$$C.B.S \Rightarrow |x-t| \leq \sqrt{L_n(x-t)^2} = \sqrt{x^2 L_n e_0 - 2xt L_n e_1 + t^2 L_n e_2}$$

↓
tinde la
0

$$\rightarrow \sqrt{1^2 - 2x^2 + x^2} = 0$$

$$|L_n f(x) - f(x)| \rightarrow 0$$

$$|f(x)(L_n e_0 - e_0) + L_n f(x) - f(x)|$$

$$\leq |L_n f - L_n e_0 f(x)| + \|f(x)\| |L_n e_0 - e_0|$$

↓
0

Demonstrație pt operator liniar de aproximație: calculăm L_0, L_1, L_2 și arătăm că tind uniform la e_0, e_1, e_2



$$B_n: C[0,1] \rightarrow \Pi_n, B_n(f) \rightarrow f$$

I₁: $f \in C[0,1] \rightarrow \mathbb{R}$ este o funcție monotona, atunci $(B_n(f))_{n \geq 1}$ are aceeași monotonicitate cu f

Dem: f - crescătoare, $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$

$$\Rightarrow B_n(f)(x_1) \leq B_n(f)(x_2) \Leftrightarrow B'_n(f)(x) \geq 0$$